Fusion ERP Lab Assignment 6B: Phase 2 Final Report

Review, Test, and Improve the AI-Generated Employee Leave Management (ELM) Module

Student Name: Sumit Kumar

Roll Number: 24BCS245 (Batch B)

Target Module: Employee Leave Management (`modules/elm`)

Assigned Questionnaire: Section S3 (30 Questions)

Git Branch: `assignment-6-elm`

PR URL: https://github.com/FusionIIIT/Fusion-Integrated/pull/388

Repository Fork: Sumit-0011/Fusion-Integrated

Final Status: 452/452 Tests Passed & Deployment Ready

1. Executive Summary

This report documents the completion of Phase 2 for Fusion ERP Lab Assignment 6B. Taking the preserved Phase-1 implementation (commit `30c7079`, tagged `phase-1-elm-original`), we conducted an exhaustive review, test, problem identification, repair, and verification process. The module was transformed from an isolated, in-memory skeleton into a fully persistent, secure, and deployment-ready ELM module integrated with the Fusion Base.

Key Verification Metrics:

- **Full Platform Backend:** 452 / 452 pytest tests passing (0 failures, 0 errors).
- **ELM Backend Suite:** 72 / 72 tests passing across domain policy, state machine, service layer, and REST API.
- **Frontend Client Suite:** 37 / 37 unit tests passing, zero TypeScript errors, and production Vite bundle built cleanly.
- **Access Control:** `manage.py permission_manifest --check` verified and matched to `registry/permissions.json` .
- **Section S3 Questionnaire:** 30 / 30 questions fully solved with 95–100% confidence and cited specification clauses.

2. Phase-1 Baseline Preservation

As required by the assignment guidelines, the original Phase-1 implementation was preserved verbatim:

- **Commit:** `30c707989ded4a2570058861d22ad0a21a8592c4`
- **Tag:** `phase-1-elm-original` (pushed to origin repository).
- **State:** Contained pure-Python leave calculation logic and basic module registration, but lacked ORM persistence, workflows, APIs, and UI.
- **Prompt Log:** Stored and accessible at `docs/phase-1-prompt-log.md` .

3. Phase-2 Problem Identification (18 Meaningful Issues)

Category	Identified Problem / Gap	Impact & Resolution
Business Rules	Missing Half-Day Casual Leave (0.5 day)	Implemented forenoon/afternoon half-day counting and validation.

Category	Identified Problem / Gap	Impact & Resolution
Business Rules	Missing Balance Accumulation Limits	Enforced 300-day cap for Earned Leave (EL) and 20-day cap for Commuted Leave (COL).
Business Rules	No Substitute Overlap Check (BR-EL-014)	Added interval collision checks preventing substitute nominee conflicts.
Business Rules	Missing Station Leave Address Enforcement	Enforced address requirements for outstation categories (EL, COL, VL).
Business Rules	No Balance Refund on Early Resumption	Calculated actual return dates and credited unused days back to user balance.
Workflows (CW-EL-01)	Missing Multi-Tier Approval Chain	Modeled Unit Head direct approval for CL/RH vs. forwarding to Sanctioning Authority.
Workflows (CW-EL-01)	No Substitute Acceptance Gate	Added `AWAITING_SUBSTITUTE` state; blocks approval until accepted.
Workflows (CW-EL-02)	Missing Cancellation Workflow	Added `CANCELLATION_PENDING` state for approved leaves prior to start date.
Workflows (CW-EL-03/04)	Missing Extension & Resumption States	Added extension requests while ongoing and duty resumption verification.
Data Persistence	No Database ORM Models	Created `LeaveRequest`, `LeaveBalance`, `LeaveTransition`, and `HolidayCalendar`.
Architecture (ADR-0013)	Cross-Module Foreign Key Prevention	Stored integer `user_id` references; no foreign keys to auth tables.
Auditability	No Immutable State Transition Log	Created append-only `LeaveTransition` table logging actor, timestamp, and reasons.
Platform Security	System vs. Role Permission Conflict	Removed `elm.admin.year_end` from role grants; kept in `SYSTEM_PERMISSIONS`.
Platform Security	Orphaned Permission (`configure_sla`)	Pruned unused permission to satisfy fail-closed manifest validation.
API Layer	Zero REST API Endpoints	Created fail-closed DRF views with `HasModuleGrant` and `HasPermission`.
Environment	Windows CP1252 Unicode Crashes	Sanitized typographic unicode box/arrow characters to clean standard ASCII.
Frontend UI	Total Absence of User Interface	Built complete Mantine React application under `client/src/modules/elm/`.
Frontend Integration	Route Registration & Type Contracts	Registered in `registry.ts` and resolved `DataTable` empty state props.

4. Implementation Architecture

4.1 Layering & Separation of Concerns

- **Pure Domain Logic (`modules/elm/domain/`):** Zero Django/ORM dependencies. `leave_policy.py` calculates continuous vs. non-continuous calendar days, rollover (2 VL : 1 EL), and overlap intervals. `state_machine.py` implements a declarative state transition table mirroring CW-EL-01 through CW-EL-04.
- **Data Layer (`modules/elm/models/` & `migrations/`):** Models with explicit DB constraints, indexing on `(user_id, status)`, and integer user IDs (ADR-0013).
- **Service Layer (`modules/elm/services/`):** Atomic transactional functions (`leave_service.py` and `balances.py`) orchestrating state transitions, balance updates, and audit trail insertions.
- **Selectors (`modules/elm/selectors/`):** Fail-closed querysets scoping records by caller role (e.g. `visible_leave_requests`, `pending_approvals`).
- **REST API (`modules/elm/api/`):** Endpoints mounted at `/api/v1/elm/` enforcing `HasModuleGrant("elm")` and granular permissions.
- **Frontend (`client/src/modules/elm/`):** Full React/Mantine client interface featuring Apply Leave, My Leaves, Balance Overview, Substitute Approvals, Administrative Offline Recording, and Holiday Calendar views.

5. Verification Results

Component / Test Suite	Command	Tests Run	Result
Full Platform Backend	`pytest`	452	452 Passed (24.22s)
ELM Backend Suite	`pytest modules/elm/tests/`	72	72 Passed (7.08s)
Access Control Suite	`pytest modules/accesscontrol/tests/`	21	21 Passed (5.83s)
Permission Manifest	`manage.py permission_manifest --check`	N/A	Manifest Valid & Matches
Frontend TypeScript	`npm run typecheck`	N/A	0 Type Errors
Frontend Unit Tests	`npm test`	37	37 Passed (4.97s)
Vite Production Build	`npm run build`	1015 modules	Built in 8.63s

6. Section S3 Questionnaire Solutions Summary

All 30 questions from `docs/questionnaires/S3.xlsx` were solved and written to sheet `S3 Student Form` with 95–100% confidence. Detailed reasoning is tracked in `docs/questionnaires/S3_SOLUTIONS.md` .

Q#	Question Focus	Ans	Conf	Primary Specification Citation
1	Change in COL carry-forward rule	**A**	100%	BR-EL-009 (Commuted leave carry-forward)
2	Resumption return cannot be verified	**B**	100%	CW-EL-04 / UC-EL-07 (Remain awaiting verification)
3	Extension skips substitute response	**D**	100%	CW-EL-03 (Substitute gate and Unit Head review)

Q#	Question Focus	Ans	Conf	Primary Specification Citation
4	Unpublished replacement calendar	**A**	100%	BR-EL-031 (Use currently effective published calendar)
5	Pre-acceptance category validation	**A**	100%	BR-EL-001 (Validate category eligibility under policy)
6	Difference: cancellation vs extension	**D**	100%	BR-EL-023 / BR-EL-025 (Prior to start vs ongoing)
7	Higher sanction without substitute routing	**B**	100%	CW-EL-01 (Unit Head -> Establishment -> Sanction Authority)
8	Continuous-count EL across holiday	**C**	100%	BR-EL-011 (Intervening holidays count as leave)
9	Verified early resumption effects	**A**	100%	BR-EL-028 / CW-EL-04 (Record actual period, restore unused days)
10	Obtain Printable Leave Application goal	**B**	100%	UC-EL-04 (Export/printable representation)
11	Purpose/reason character limit	**E**	100%	FR (Cannot be determined from specification)
12	Nominee with pending leave eligibility	**C**	100%	BR-EL-014 (Overlapping pending leave is ineligible)
13	Rejected cancellation original state	**D**	100%	BR-EL-023 / CW-EL-02 (Remains Approved — Unchanged)
14	Staff EL vs Faculty VL contrast	**C**	100%	BR-EL-005–008 (Staff 30 EL; Faculty 60 VL + 2:1 rollover)
15	Valid CL path with substitute	**D**	100%	CW-EL-01 (Draft -> Awaiting Substitute -> Unit Head -> Rejected)
16	Record offline sanctioned leave boundary	**B**	100%	BR-EL-029 (Records pre-approved leave; no second approval)
17	Exact final notification channel	**E**	100%	FR (Cannot be determined from specification)
18	Year-end treatment of unused RH	**C**	100%	BR-EL-003 (Lapses; does not carry forward)
19	Additional time on ongoing COL	**B**	100%	CW-EL-03 / UC-EL-009 (Submit Leave Extension)
20	Excluded from continuous-day counting	**C**	100%	BR-EL-010 (CL, SCL, and RH exclude weekends/holidays)
21	Decline loop limit before auto-rejection	**E**	100%	BR-EL-015 (Cannot be determined from specification)
22	Effective hierarchy policy routing	**C**	95%	BR-EL-030 (Use currently effective published policy)

Q#	Question Focus	Ans	Conf	Primary Specification Citation
23	Rejected extension resulting state	**D**	100%	CW-EL-03 (Ongoing Leave — Original Unchanged)
24	Higher-sanction lifecycle progression	**D**	100%	CW-EL-01 (Submission -> Unit Head -> Establishment -> Sanction)
25	Two overlapping full-day leaves	**A**	100%	BR-EL-012 (Detect and prevent impermissible overlap)
26	Mandatory medical certificate template	**E**	100%	BR-EL-009 (Cannot be determined from specification)
27	Configurable SLA thresholds rule	**B**	100%	BR-EL-037 / CFR-EL-023 (Configurable reminder thresholds)
28	Validation failure before submission	**B**	100%	UC-EL-001 (Request does not progress until corrected)
29	Substitute data change removed step	**A**	100%	CW-EL-01 (Fresh substitute response requirement)
30	Extension balance posting rule	**A**	100%	BR-EL-025 / CW-EL-03 (Post only incremental additional days)

7. Artifact & Evidence Links

- **Pull Request (PR #388):** https://github.com/FusionIIIT/Fusion-Integrated/pull/388
- **Improved ELM Code Folder:** Sumit-0011/Fusion-Integrated/tree/assignment-6-elm/modules/elm
- **Phase-1 Preserved Tag:** Sumit-0011/Fusion-Integrated/tree/phase-1-elm-original/modules/elm
- **Phase-1 Prompt Log:** docs/phase-1-prompt-log.md
- **Phase-2 Prompt Log:** docs/phase-2-prompt-log.md
- **Backend Test Suite:** modules/elm/tests/
- **Questionnaire S3 Solutions:** docs/questionnaires/S3_SOLUTIONS.md